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Data Science

PANDAS

What is Pandas?

- Pandas is a Python library used to work with data. It helps us:
- Store data
- Read data
- Analyze data
- Modify data

In [186...

```
import pandas as pd
```

Pandas Series

What is a Series?

- A Series is a one-dimensional array with labels (indexes).
- It stores a single column of data.

Example:

Index	Value
0	10
1	20
2	30
3	40

Creating a Series

Series From a List

```
In [191... numbers = [10, 20, 30, 40]

num_series = pd.Series(numbers)

print(num_series)
```

```
0    10
1    20
2    30
3    40
dtype: int64
```

Series with Custom Index

Creating a Series of Student Marks

```
In [193... marks = [85, 90, 75]

mark_series = pd.Series(marks, index=["Rahul", "Anu", "John"])

print(mark_series)
```

```
Rahul    85
Anu      90
John     75
dtype: int64
```

Creating a Series from a Dictionary

```
In [195... student = {
    "Rahul": 85,
    "Anu": 90,
    "John": 75
}

student_series = pd.Series(student)

print(student_series)
```

```
Rahul    85
Anu      90
John     75
dtype: int64
```

Accessing Values

By Position

```
In [198... print(mark_series.iloc[0])
```

```
85
```

By Label

```
In [200... print(mark_series["Rahul"])
```

85

Series Properties

Index

```
In [203... print(mark_series.index)
```

Index(['Rahul', 'Anu', 'John'], dtype='str')

Values

```
In [205... print(mark_series.values)
```

[85 90 75]

Data Type

```
In [207... print(mark_series.dtype)
```

int64

Functions

Sum

```
In [210... print(mark_series.sum())
```

250

Average

```
In [212... print(mark_series.mean())
```

83.33333333333333

Maximum

```
In [214... print(mark_series.max())
```

90

Minimum

```
In [216... print(mark_series.min())
```

75

Count

```
In [218... print(mark_series.count())
```

3

Head()

Shows the first values.

```
In [220... print(mark_series.head())
```

```
Rahul    85  
Anu      90  
John     75  
dtype: int64
```

```
In [221... print(mark_series.head(2))
```

```
Rahul    85  
Anu      90  
dtype: int64
```

Tail()

Shows the last values.

```
In [223... print(mark_series.tail())
```

```
Rahul    85  
Anu      90  
John     75  
dtype: int64
```

```
In [224... print(mark_series.tail(2))
```

```
Anu      90  
John     75  
dtype: int64
```

Filtering Data

Display values greater than 20.

```
In [226... import pandas as pd  
  
a = pd.Series([10, 20, 30, 40, 50])
```

```
print(a[a > 20])
```

```
2    30
3    40
4    50
dtype: int64
```

Updating a Value

In [228...

```
a[1] = 100

print(s)
```

```
0    10
1    20
2    30
3    40
dtype: int64
```

Adding a New Value

In [230...

```
a["New"] = 60

print(s)
```

```
0    10
1    20
2    30
3    40
dtype: int64
```

Deleting a Value

In [232...

```
a = a.drop("New")

print(s)
```

```
0    10
1    20
2    30
3    40
dtype: int64
```

Pandas DataFrame

What is a DataFrame?

- A DataFrame is a two-dimensional table in Pandas.
- It contains rows and columns, just like an Excel spreadsheet.

Example:

Name	Age	Marks
Rahul	20	85
Anu	21	90
John	19	78

Creating a DataFrame

```
In [236... import pandas as pd

data = {
    "Name": ["Rahul", "Anu", "John"],
    "Age": [20, 21, 19],
    "Marks": [85, 90, 78]
}

df = pd.DataFrame(data)

print(df)
```

	Name	Age	Marks
0	Rahul	20	85
1	Anu	21	90
2	John	19	78

Viewing the Data

```
In [238... print(df)
```

	Name	Age	Marks
0	Rahul	20	85
1	Anu	21	90
2	John	19	78

First 5 Rows

```
In [240... print(df.head())
```

	Name	Age	Marks
0	Rahul	20	85
1	Anu	21	90
2	John	19	78

```
In [241... print(df.head(2))
```

	Name	Age	Marks
0	Rahul	20	85
1	Anu	21	90

Last 5 Rows

```
In [243... print(df.tail())
```

	Name	Age	Marks
0	Rahul	20	85
1	Anu	21	90
2	John	19	78

```
In [244... print(df.tail(2))
```

	Name	Age	Marks
1	Anu	21	90
2	John	19	78

Information About Data

Shape (Rows, Columns)

```
In [247... print(df.shape)
```

(3, 3)

Column Names

```
In [249... print(df.columns)
```

Index(['Name', 'Age', 'Marks'], dtype='str')

Data Types

```
In [251... print(df.dtypes)
```

Name str
Age int64
Marks int64
dtype: object

Information About the DataFrame

```
In [253... print(df.info())
```

```
<class 'pandas.DataFrame'>  
RangeIndex: 3 entries, 0 to 2  
Data columns (total 3 columns):  
#   Column  Non-Null Count  Dtype  
---  -  
0   Name    3 non-null      str  
1   Age     3 non-null      int64  
2   Marks   3 non-null      int64  
dtypes: int64(2), str(1)  
memory usage: 216.0 bytes  
None
```

Statistical Information

In [255...] `df.describe()`

Out[255...]

	Age	Marks
count	3.0	3.000000
mean	20.0	84.333333
std	1.0	6.027714
min	19.0	78.000000
25%	19.5	81.500000
50%	20.0	85.000000
75%	20.5	87.500000
max	21.0	90.000000

Selecting Data

Select One Column

In [257...] `print(df["Name"])`

```
0    Rahul
1     Anu
2     John
Name: Name, dtype: str
```

Select Multiple Columns

In [259...] `print(df[["Name", "Marks"]])`

```
   Name  Marks
0  Rahul    85
1   Anu    90
2   John    78
```

Accessing Rows

Using loc (Label)

In [261...] `print(df.loc[0])`

```
Name    Rahul
Age      20
Marks    85
Name: 0, dtype: object
```

Using iloc (Position)

```
In [263... print(df.iloc[1])
```

```
Name    Anu
Age      21
Marks    90
Name: 1, dtype: object
```

Functions

sum

```
In [266... print(df["Marks"].sum())
```

```
253
```

Average

```
In [268... print(df["Marks"].mean())
```

```
84.33333333333333
```

Maximum

```
In [270... print(df["Marks"].max())
```

```
90
```

Minimum

```
In [272... print(df["Marks"].min())
```

```
78
```

Filtering Data

Display students with marks greater than 80.

```
In [274... print(df[df["Marks"] > 80])
```

```
   Name  Age  Marks
0  Rahul   20    85
1   Anu   21    90
```

Adding a New Column

```
In [276... df["Grade"] = ["B", "A", "C"]  
  
print(df)
```

	Name	Age	Marks	Grade
0	Rahul	20	85	B
1	Anu	21	90	A
2	John	19	78	C

Deleting a Column

Note: axis=1 means delete a column. (axis=0 is for rows.)

```
In [279... df = df.drop("Grade", axis=1)  
  
print(df)
```

	Name	Age	Marks
0	Rahul	20	85
1	Anu	21	90
2	John	19	78

Saving a DataFrame

```
In [281... df.to_csv("students.csv", index=False)
```